Oct 27

$$\sum_{n=2}^{\infty} (-1)^n \underbrace{(3)^{\frac{1}{n}}}_{a_n}$$

Try AST

(1) $a_n \geq a_{n+1}$ ✓

(2) $\lim_{n\to\infty} a_n = 0$ ✗

$3^{\frac{1}{n}} \ ? \ 3^{\frac{1}{n+1}}$ ← This exponent ↳ smaller

$\frac{1}{n} \geq \frac{1}{n+1}$

$3^{\frac{1}{n}} \geq 3^{\frac{1}{n+1}}$

$\lim_{n\to\infty} 3^{\frac{1}{n}} = 3^0 = 1$

$\lim_{n\to\infty} 3^{\frac{1}{n}} = e^{\lim_{n\to\infty} \frac{1}{n}\ln(3)}$

$= e^0 = 1$

Try DT

if $\lim_{n\to\infty} a_n \neq 0$, it diverges

↳ entire thing, not AST's a_n

$\lim_{n\to\infty} (-1)^n (3)^{\frac{1}{n}}$

$\lim_{n\to\infty} (-1)^n \, 3^0$

when evaluating $\lim_{n\to\infty}$ and there are diff parts dependent on that tell you diff things abt conv. and div then you should use logarithms

$= \lim_{n\to\infty} (-1)^n$

↳ DNE because alternating $\neq 0$

Diverges

Diverges by divergence test

$\lim_{n\to\infty} \left(1+\frac{3}{n}\right)^n = \underbrace{e^3}_{e^3 \neq 0}$ so $\sum_{n=2}^{\infty} \left(1+\frac{3}{n}\right)^n$ diverges

↑ under divergence test

Divergence test is what you should look at first to eliminate other possibilites → Ratio and root test can tell you to use a different test if you get 1

Ratio test is a good 2nd choice if you rule out divergence